

Cash Management

Working Capital Profile

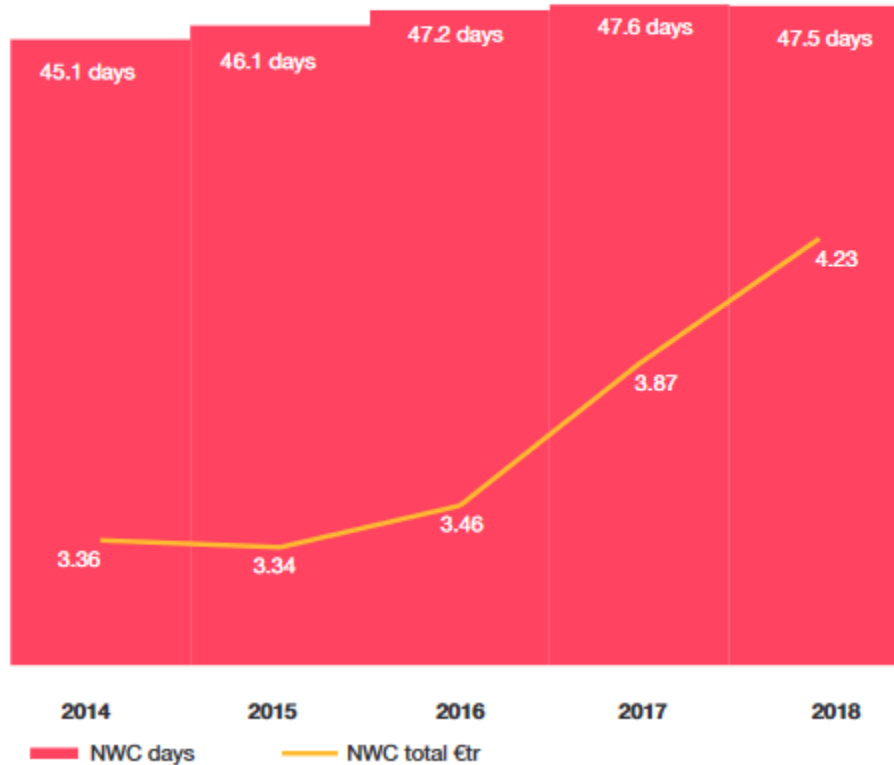
- Financial performance of the largest global listed companies in the past five years:
- Working Capital as a value driver
 - Improvements in returns have mostly come through EBIT. Some of the value created has, however, been offset by stalling net working capital (NWC) performance, restraining the improvement in return on invested capital (ROIC). Addressing excess working capital would lift overall ROIC by up to 30bps (basis points).
- Improving Working Capital
 - While net working capital increased by €360bn in 2018 (up 9.4% on 2017), relative performance in terms of days has improved marginally by 0.1 days.
- Payables Days have been unsustainable
 - For the second successive year we have seen a decrease in Days Payable Outstanding (DPO), underlining that the use of DPO as a quick fix is not sustainable in the long term as a working capital management strategy

Working Capital Commitments: Survey

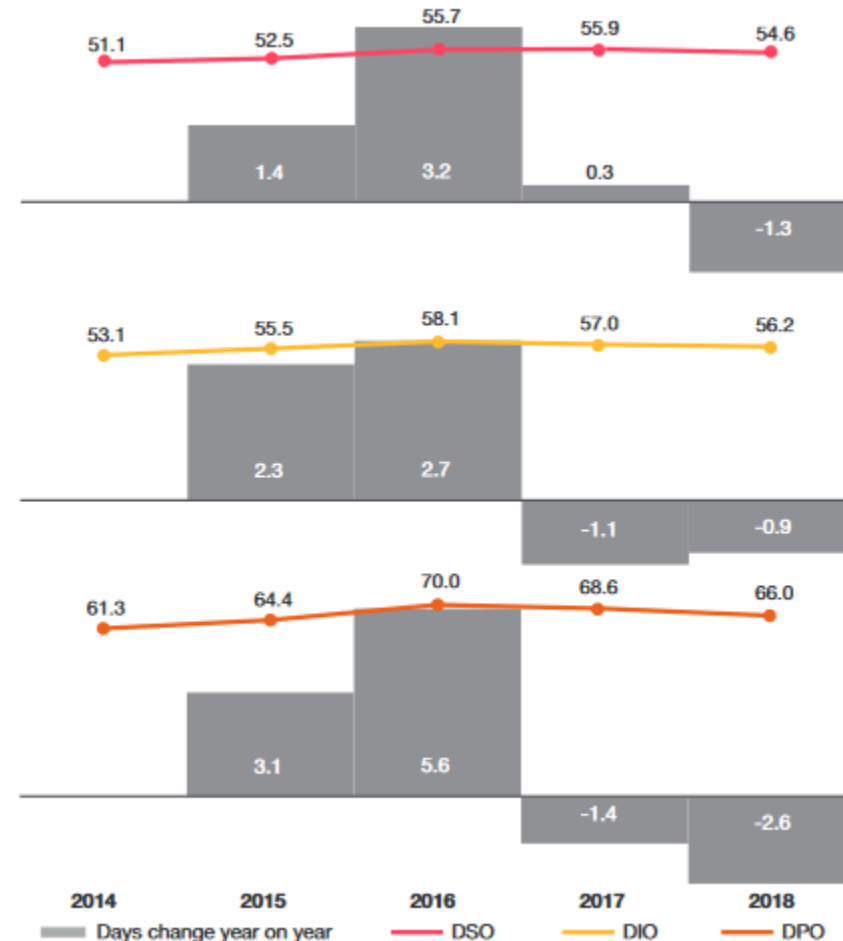
- Receivables and Inventory are major sources of opportunity
 - Many companies have finally started to achieve significant improvements in both “days sales outstanding (DSO) and “days inventory outstanding” (DIO). This has been much needed in light of the downward pressure on DPO. DSO has shown its first improvement in five years as companies have begun to address the asset side of the balance sheet.
- The need for cash is increasing
 - While revenues are up 10% over, in 2017, in 2018 operating cash flows (OCF) have declined as a proportion of sales. Companies are facing operational challenges in converting revenue into cash. During the same period, capital expenditure (CAPEX), as a percentage of revenues has continued to decline, which could suggest that companies are managing cash levels by limiting investment.

Working Capital Trend

Net working capital and working capital days



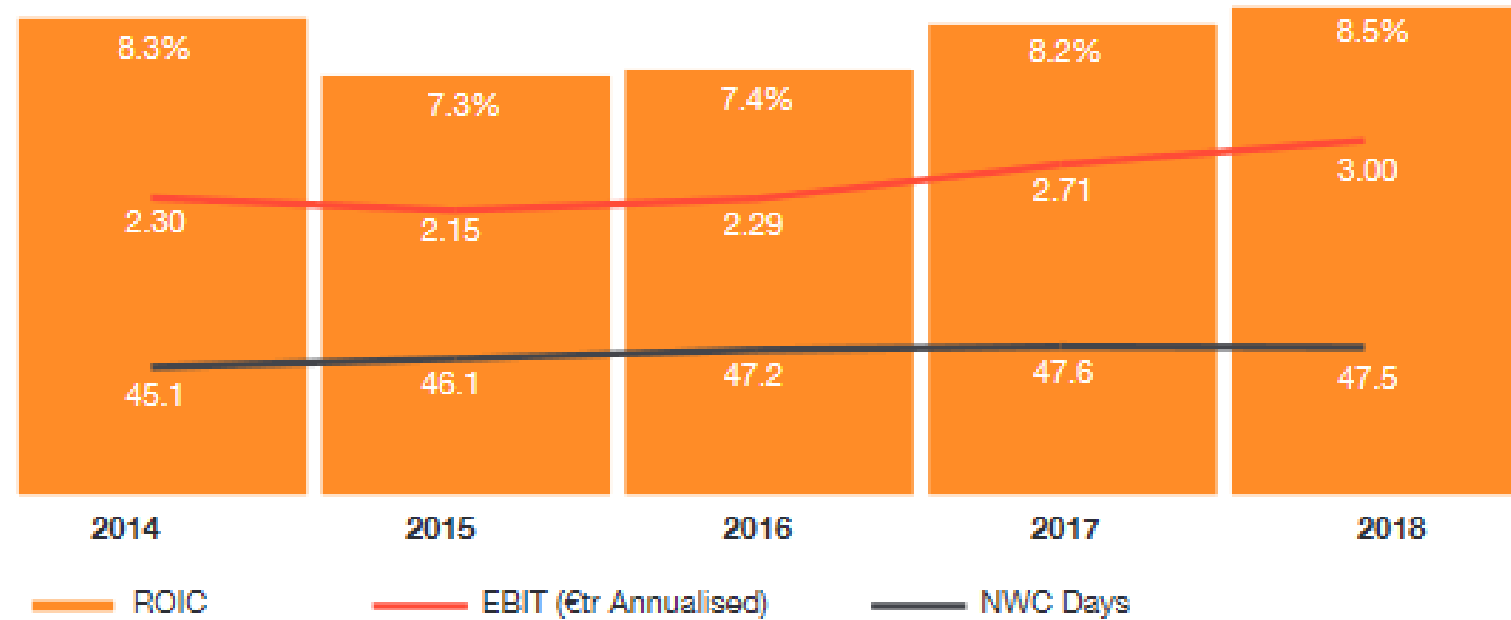
DSO, DIO and DPO trend



Source: <https://www.pwc.com/>

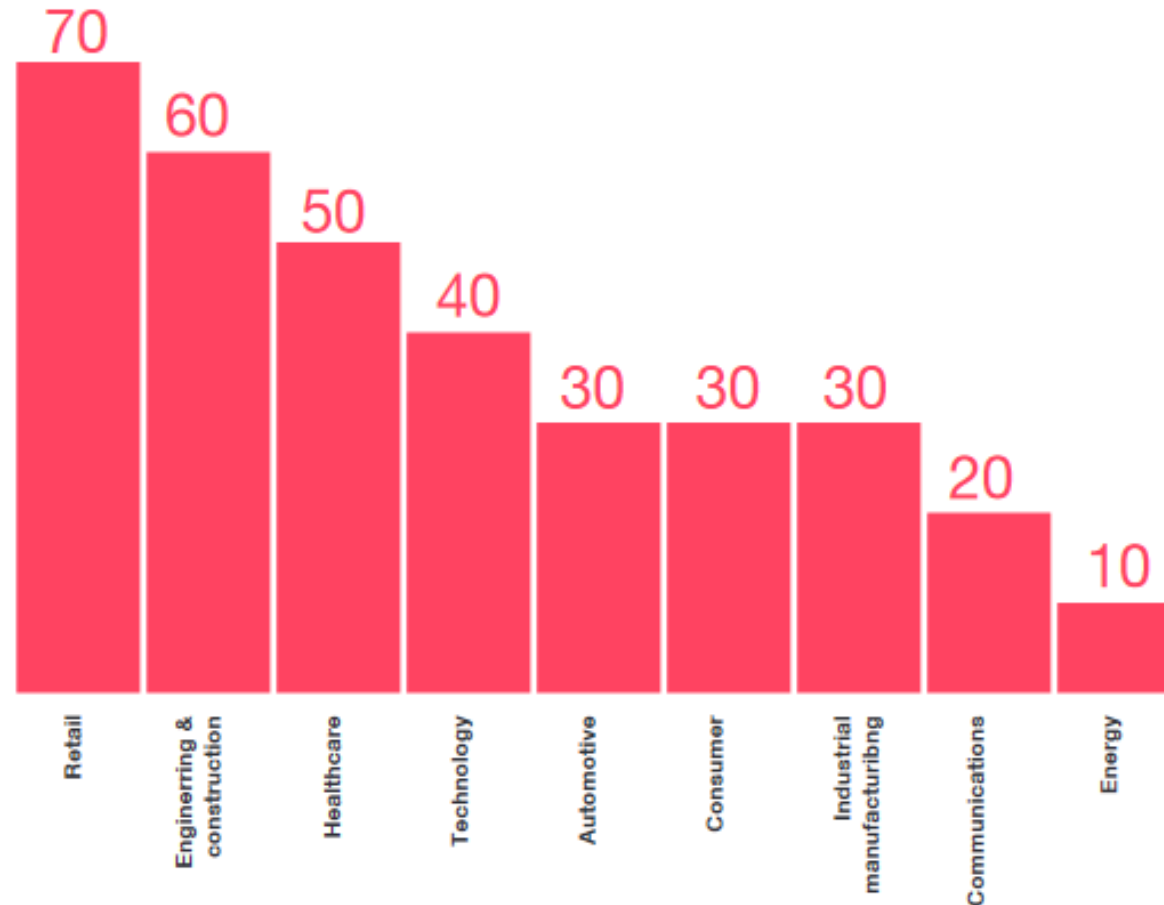
Value Driver

EBIT/ROIC trend



Basis Points Contribution

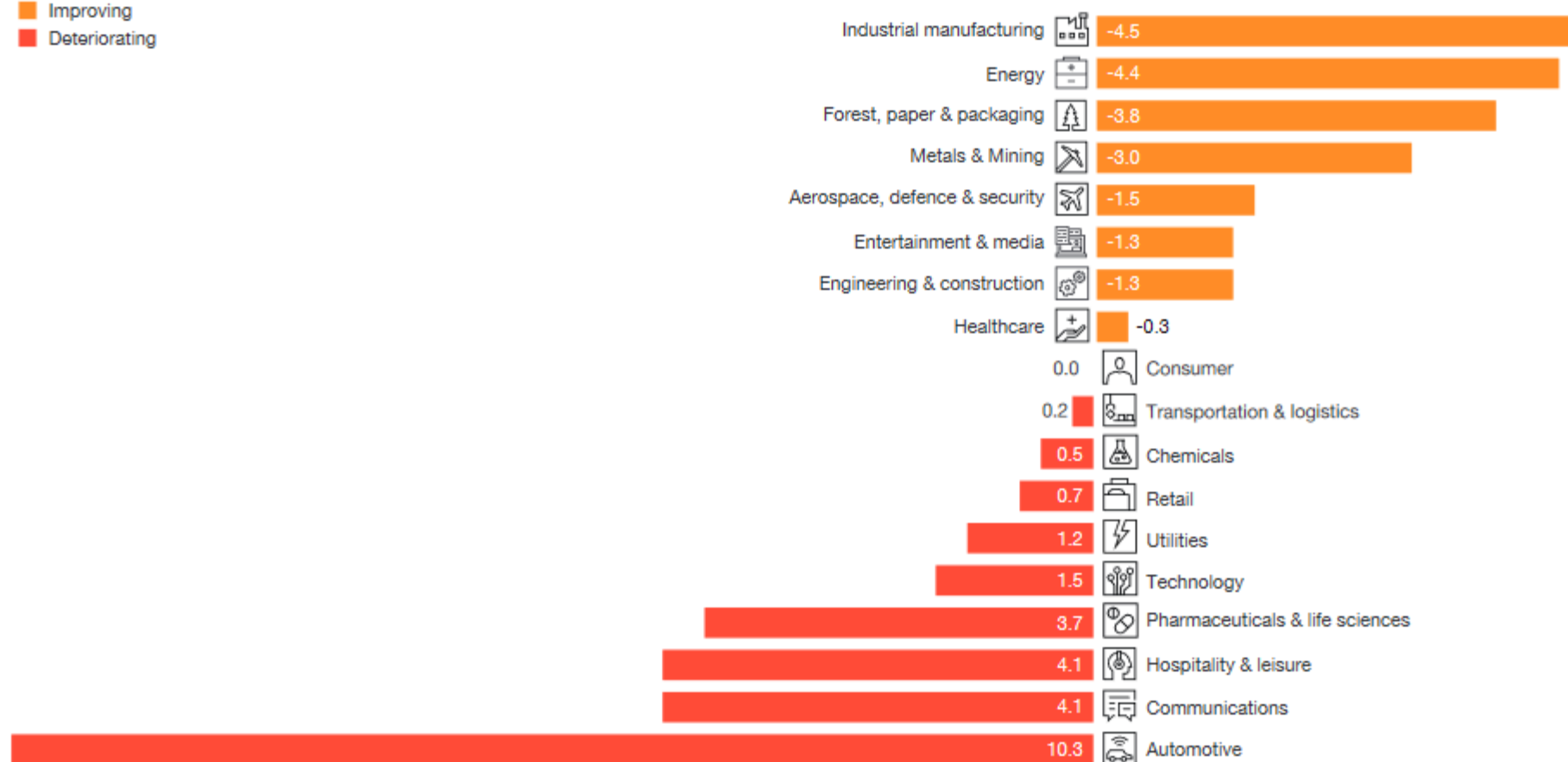
BPS improvement in ROIC through eliminating excess Working Capital



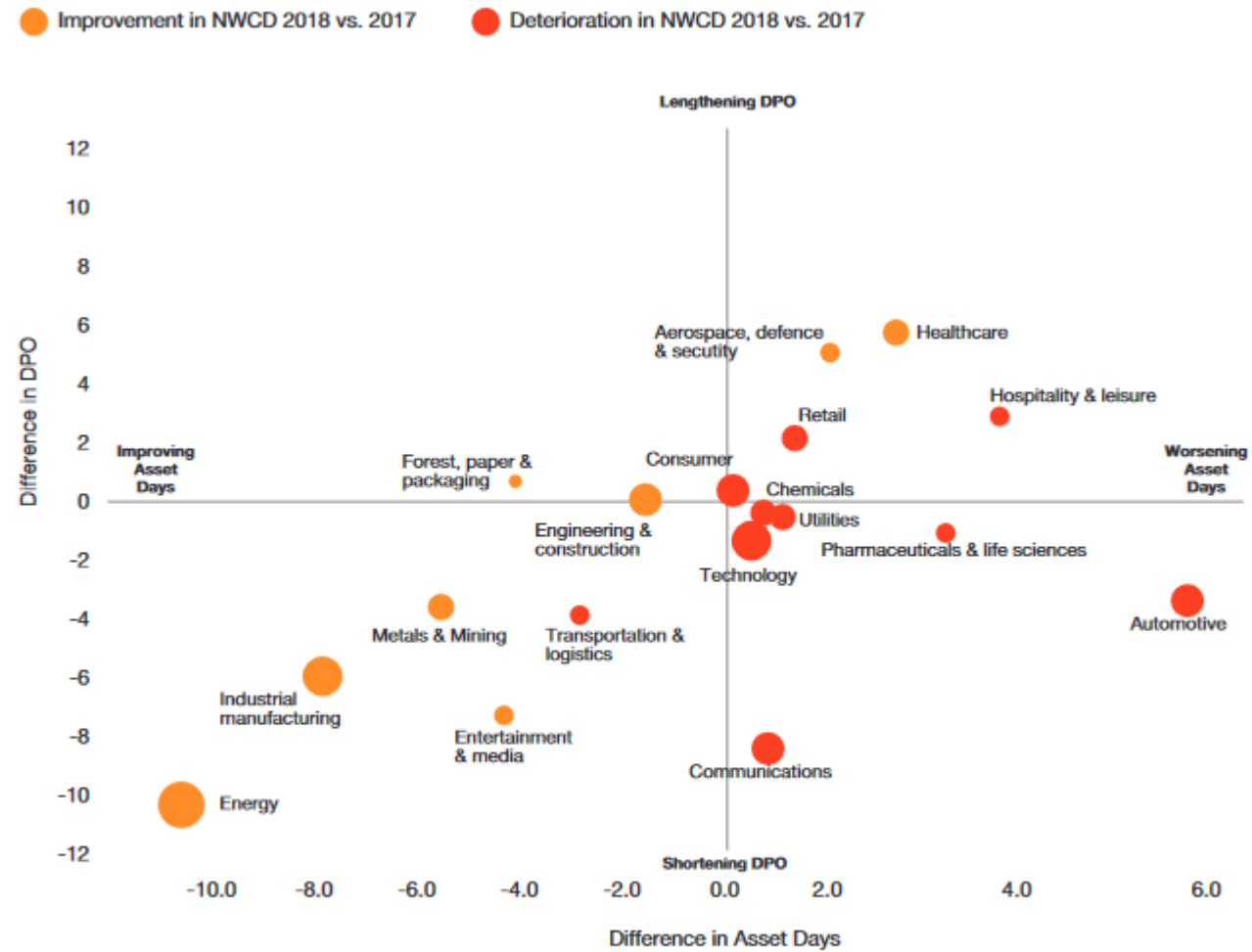
Source: <https://www.pwc.com/>

Working capital change 2017-2018 expressed in days

Improving
Deteriorating

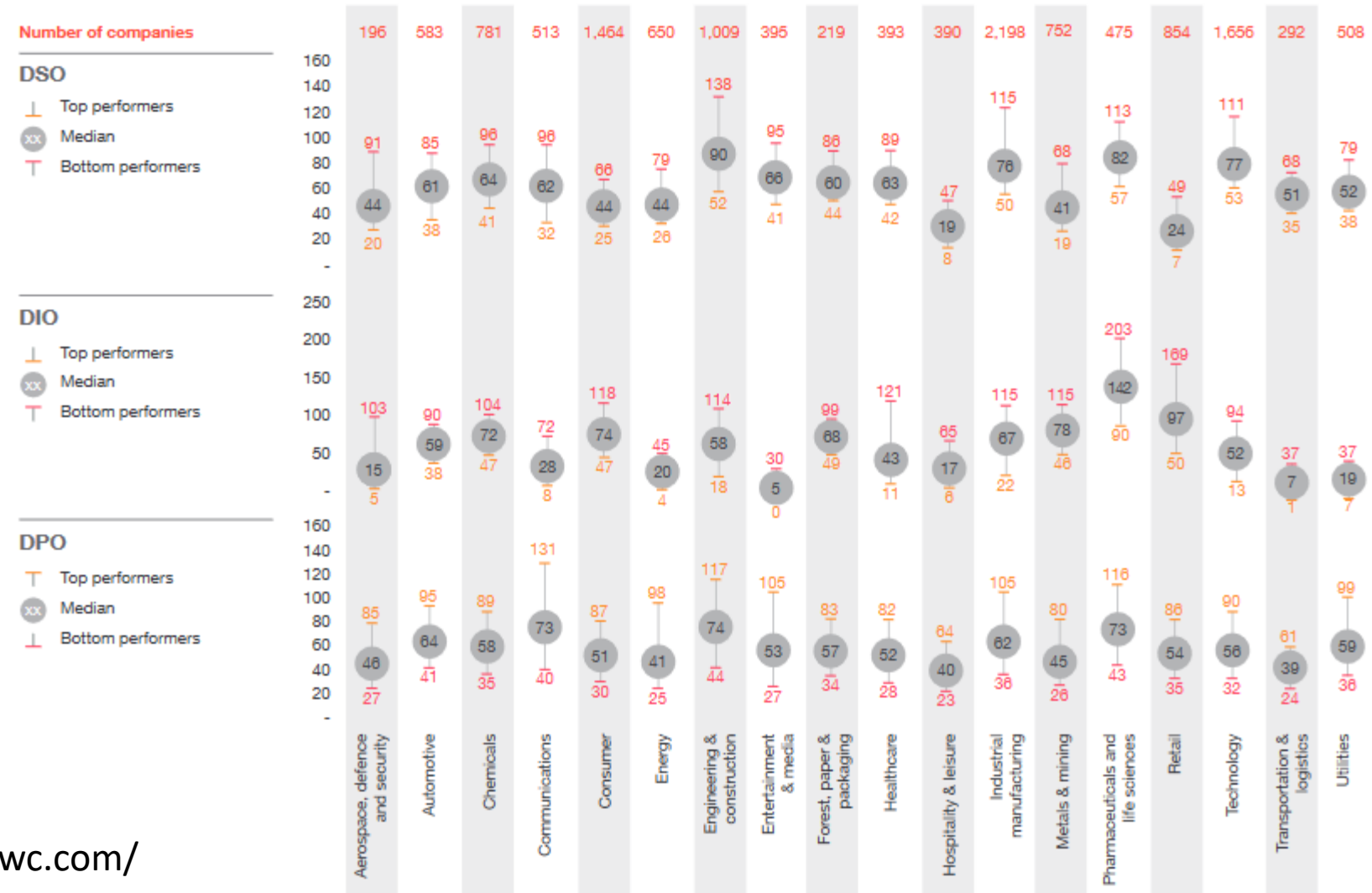


Sectoral Comparison



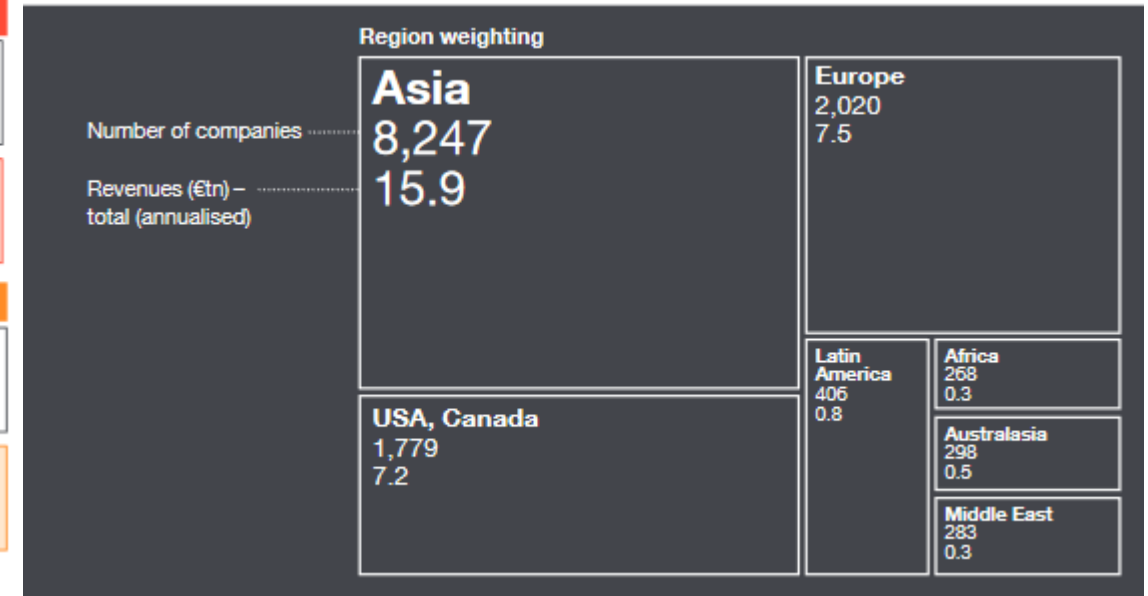
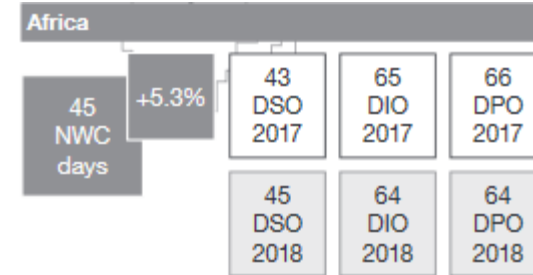
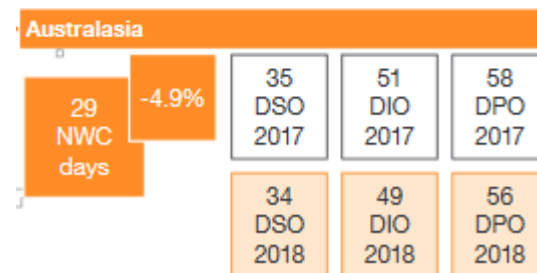
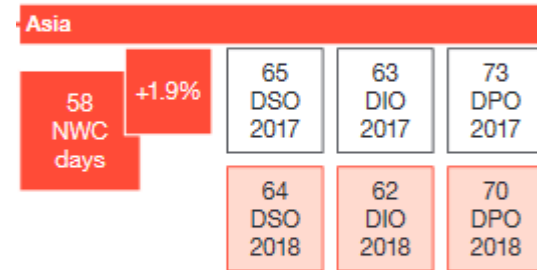
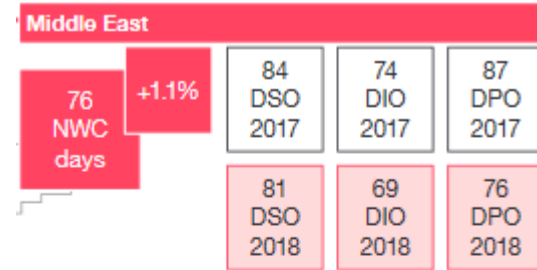
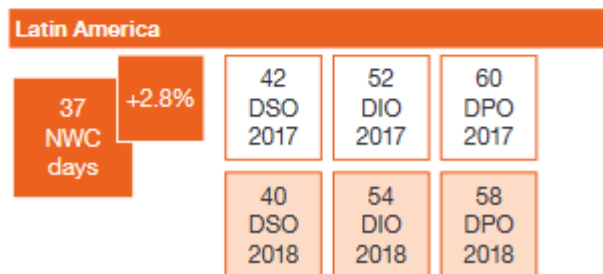
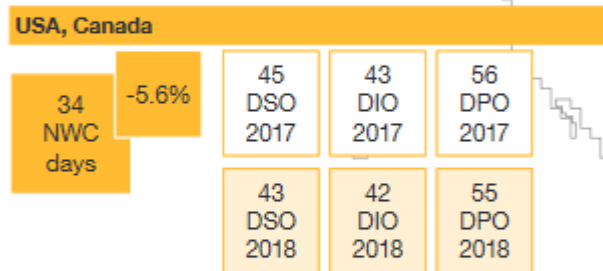
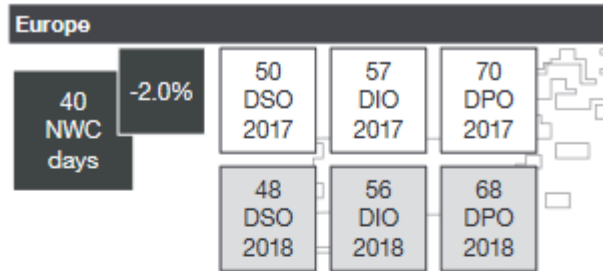
Asset days = (trade receivables and inventory)/revenue *365

Comparison with Best Working Capital Performers



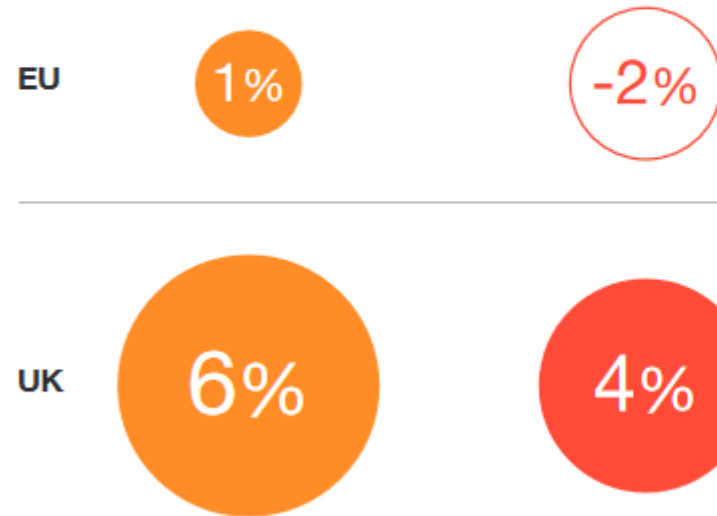
Source: <https://www.pwc.com/>

Working Capital Profile



Value Preservations

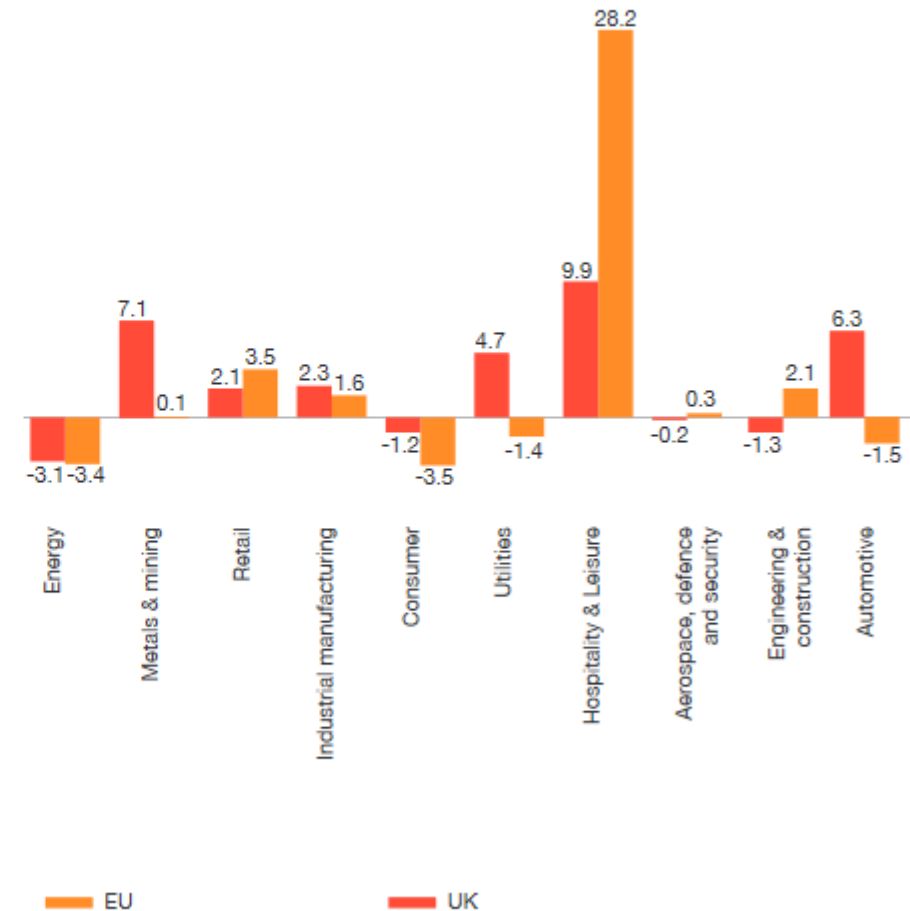
Rate of change in Net Working Capital days



Per annum percentage change in NWC days since 2014

Per annum percentage change in NWC days since 2016

EU vs UK per annum percentage change in Net Working Capital days from 2014 to 2018

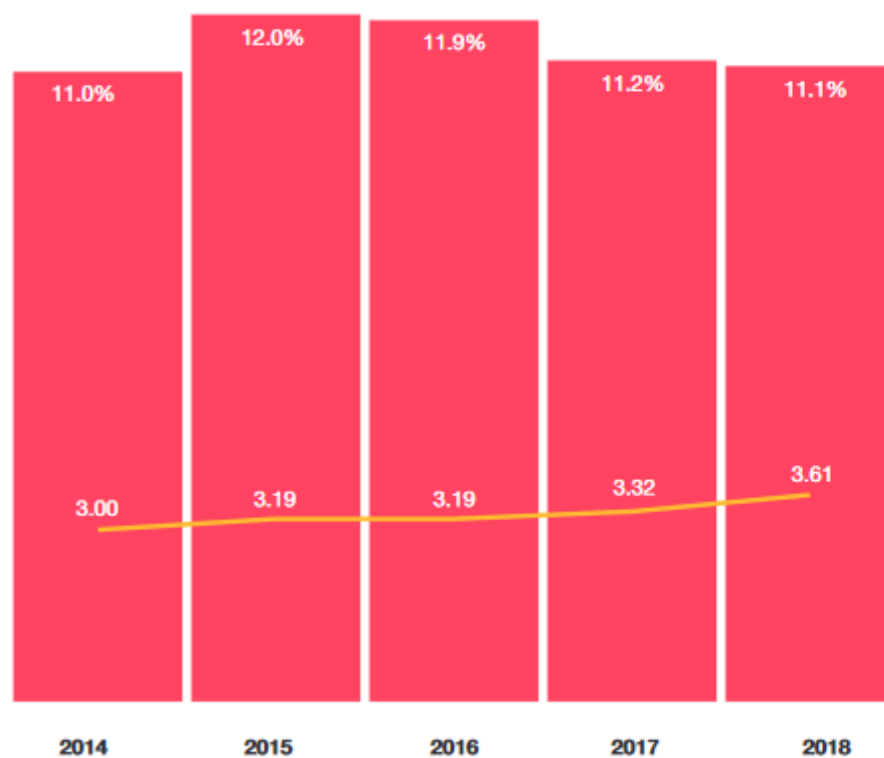


EU

UK

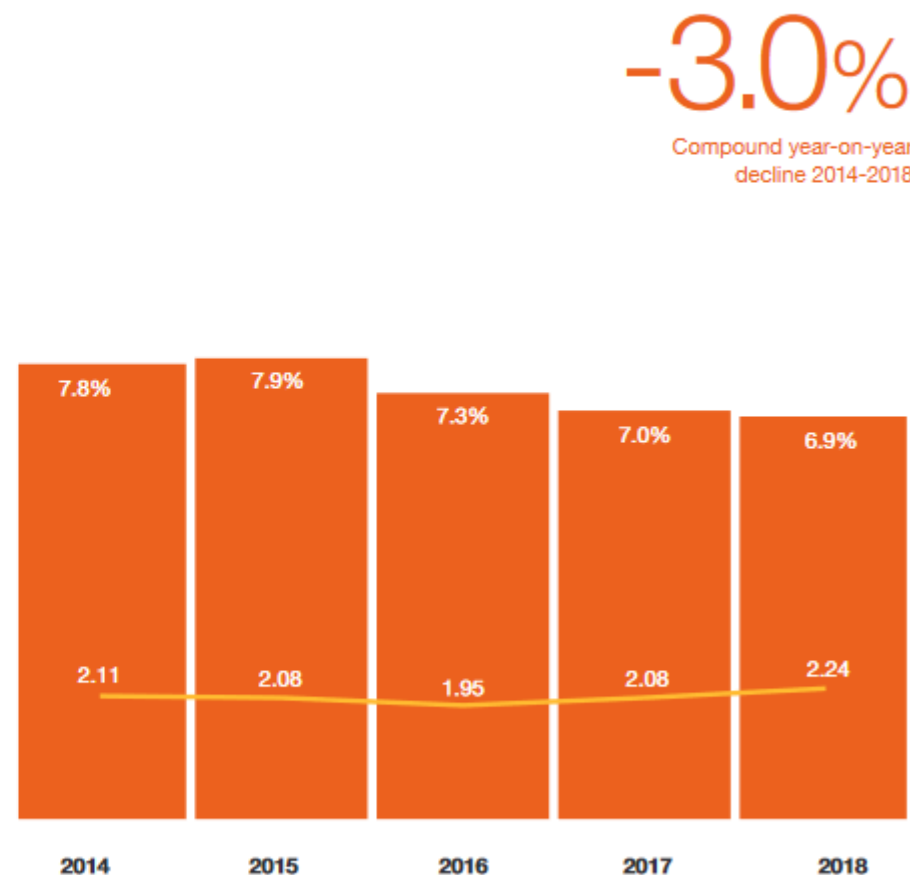
Need for Cash

Operating cash flow/revenue trend



Ops CF/revenue % Cash from ops (€tr)

CAPEX/revenue trend

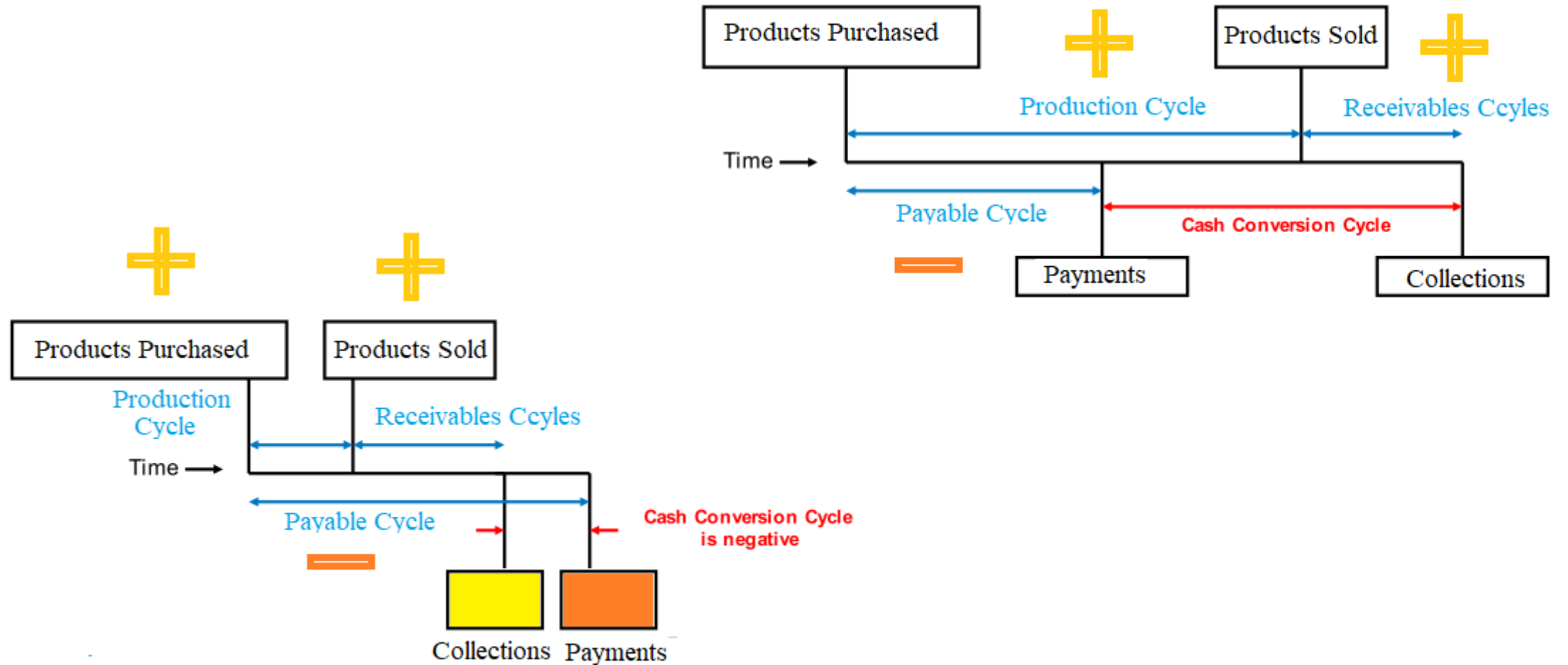


CAPEX/revenue % CAPEX €tr

Cash Conversion Cycle

- *Production cycle*: the time it takes to build and sell the product
 - begins when a customer places an order and ends when the product is shipped out.
- *Collection cycle*: the time it takes to collect from customers (collecting accounts receivable on credit sales)
 - begins when the order is filled and ends when payment is received.
- *Payment cycle*: the time it takes to pay for supplies and labor, (paying your accounts payable).
 - begins when labor is associated, or raw materials are received to start production and ends when the firm pays for purchases, raw materials and other production costs.
- Firms pay for production costs before they receive payment from their customers, a longer cash conversion cycle would tie up their finances

Cash Conversion Cycle



Cash Conversion Cycle

- Receivables Cycle + Production Cycle – Payables Cycle
- Operating Cycle - Payables Cycle
- **Production Cycle = $365 / \text{Inventory Turnover Rate}$**
 - **Inventory Turnover Rate = $\text{Cost of goods sold} / \text{Average inventory}$**
 - **Average Inventory = $(\text{Beginning inventory} + \text{ending inventory}) / 2$**
- **Receivables Cycle = $365 / \text{Receivables Turnover Rate}$**
 - **Receivables Turnover Rate = $\text{Credit sales} / \text{Average Receivables}$**
 - **Average Receivables = $(\text{Beginning receivables} + \text{ending receivables}) / 2$**
- **Payables Cycle = $365 / \text{Payables Turnover Rate}$**
 - **Payables Turnover Rate = $\text{Cost of goods sold} / \text{Average payables}$**
 - **Average Payables = $(\text{Beginning payables} + \text{ending payables}) / 2$**

Cash Conversion Cycle

- Firm can shorten its production cycle or its collection cycle, or both, while keeping its payment cycle constant or lengthened
 - Shortens the number of days to finance its operations thereby reducing its financing costs and increasing its profits.
- Focus on Receivables, Payables, and inventory
- Cash Conversion Cycle is the amount of time (in days) that a firm takes to sell inventory, collect receivables and pay accounts payable
- Combined cycle indicates how much cash is tied up in the firm's operations (procurement, production, sales, etc.)

Example

- Cash sales 200,000; Credit sales 600,000;
- Cost of goods sold 640,000.

	Balance (Beginning)	Balance (Ending)
Receivables	36000	40000
Payables	5000	9000
Inventory	6000	10000

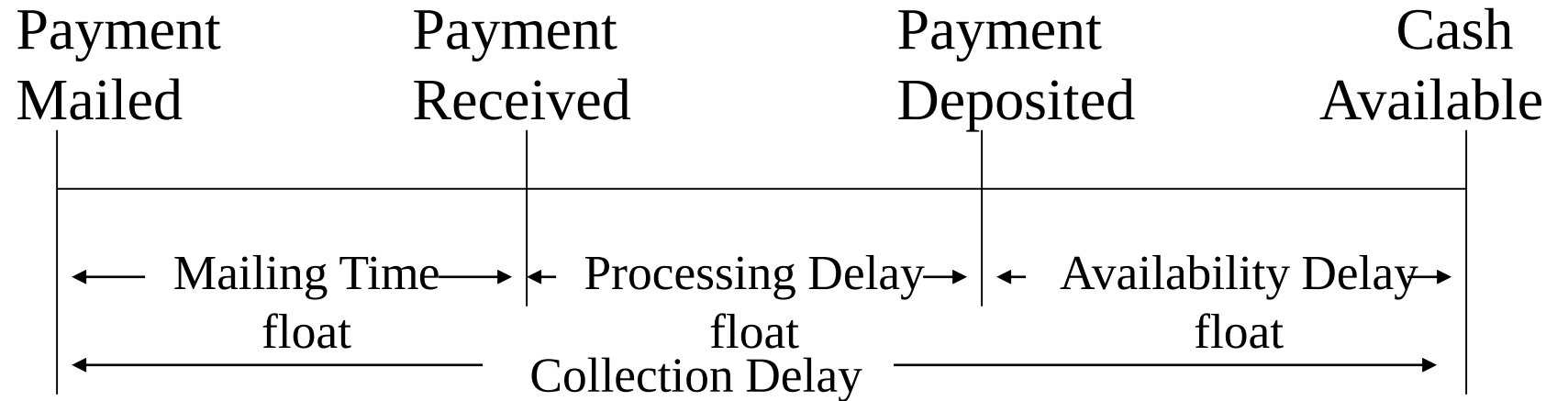
Average Receivables	Average Payables	Average Inventory
38000	7000	8000
Receivables Turnover	Credit Sales/Average receivables	15.79 (600000/38000)
Payables Turnover	Cost of Goods Sold/Average payables	91.43 (640000/7000)
Average Inventory	Cost of Goods Sold/Average inventory	80 (640000/8000)
Collection Cycle	Payment Cycle	Production Cycle
23.12 days (365/15.79)	3.99 days (365/91.43)	4.56 days (365/80)

- Operative cycle = Collection cycle – Production cycle = 27.65 days (23.12 – 4.56)
- Cash flow cycle = Operating cycle – Payable cycle = 23.69 days (27.65- 3.99)

Cash Holding

- Speculative motive: take advantage of unexpected opportunities
- Precautionary motive: meet emergency requirements
- Transaction motive: pay the day-to-day expenditure
- Compensating Balance Requirements: cash balances kept with banks to compensate for banking services
- Optimal decision of holding cash is the trade-off between
 - *Carrying costs: Opportunity costs of holding cash instead of some other income-producing asset.* **Versus**
 - *Shortage costs: Cost of not having cash or having to rapidly get the cash.*
 - *Ability to borrow marketable securities.*
 - *Scale economies in cash management.*

Float



- Float: difference between cash balance recorded in the cash account and the cash balance recorded at the bank
- Disbursement float
 - Generated when a firm writes checks
 - Available balance at bank – book balance > 0
- Collection float
 - Checks received increase book balance before the bank credits the account
 - Available balance at bank – book balance < 0
- Net float = disbursement float + collection float
- Electronic clearance systems reduce the float to 0 for a minimal charge.

Float: Example

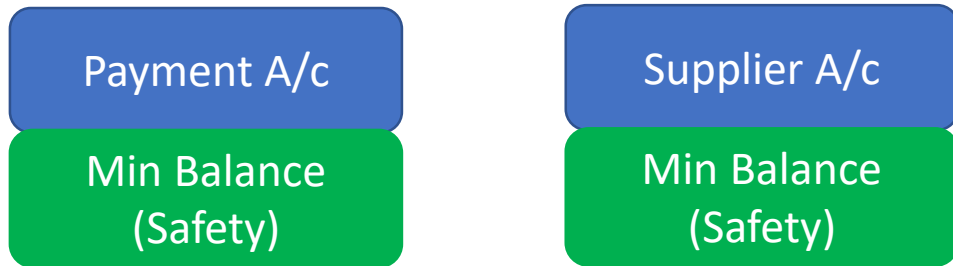
- Size of float depends on the value and the time delay
- Delay = mailing time + processing delay + availability delay
- Assume a check for 1,000 takes 3 days to reach its destination, 1 day to process, and 1 day before the bank makes the cash available
- Average daily float (assuming 30-day months) $[(3+1+1)/30] \times (1,000) = 166.67$
- Cost of float: opportunity cost of not being able to use the money
- Assume average daily float is 3 million with a weighted average delay of 5 days.
- Amount unavailable to earn interest is 15 million (5 x 3 million)
- NPV of a project that could reduce the delay by 3 days if the cost is 8 million = 1 million [cash flow is 9 million (3 x 3 million) – cost 8 million]

Float: Example

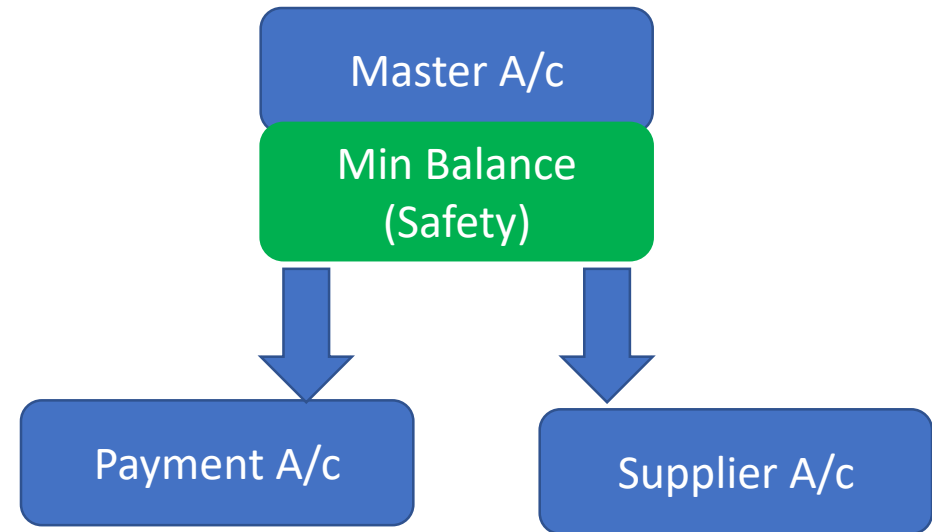
- Each business day, on average, a firm writes checks totaling 25,000 to pay its suppliers. The usual clearing time for the cheques is four days. Firm receives payments from its customers each day, in the form of cheques, totaling 40,000. The cash from the payments is available to the firm after two days.
- Disbursement float = $4 \times 25,000 = 100,000$
- Collection float = $2 \times 40,000 = 80,000$
- Net float = 20,000 ($100,000 - 80,000$) (Amount available for investment)
- If collected funds are made available faster (in one day), then the net float will be 60,000 ($100,000 - 40,000$)
- If disbursement delay is also reduced to 2 days, then the net float will be 10,000 ($50,000 - 40,000$)

Controlling Disbursements

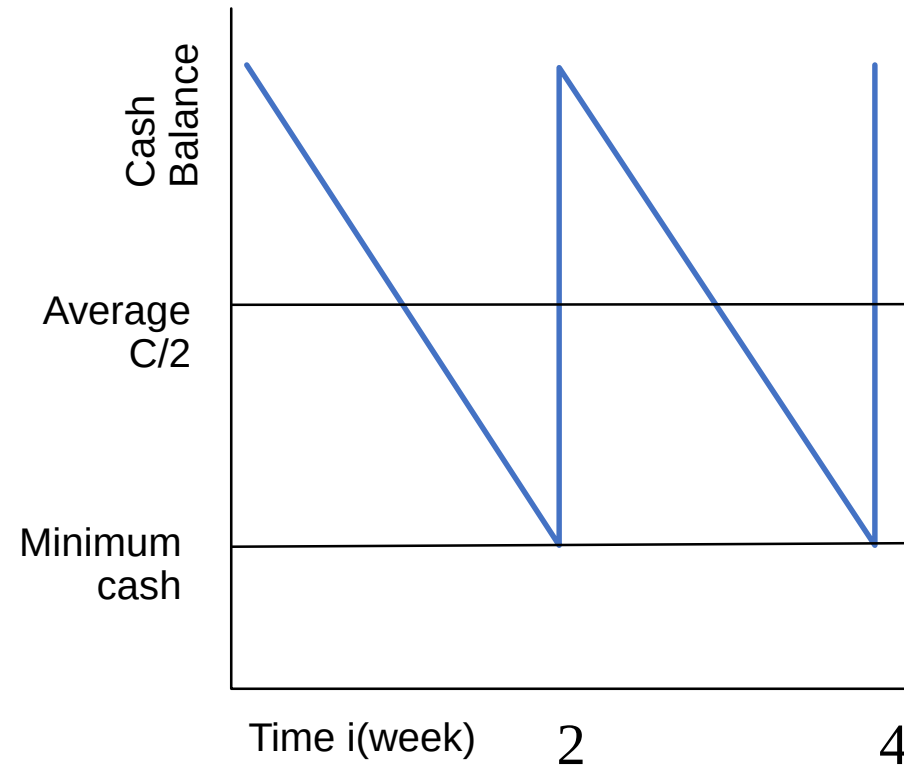
No-Zero Balance Account



Zero Balance Account



Cash Requirements for a Firm

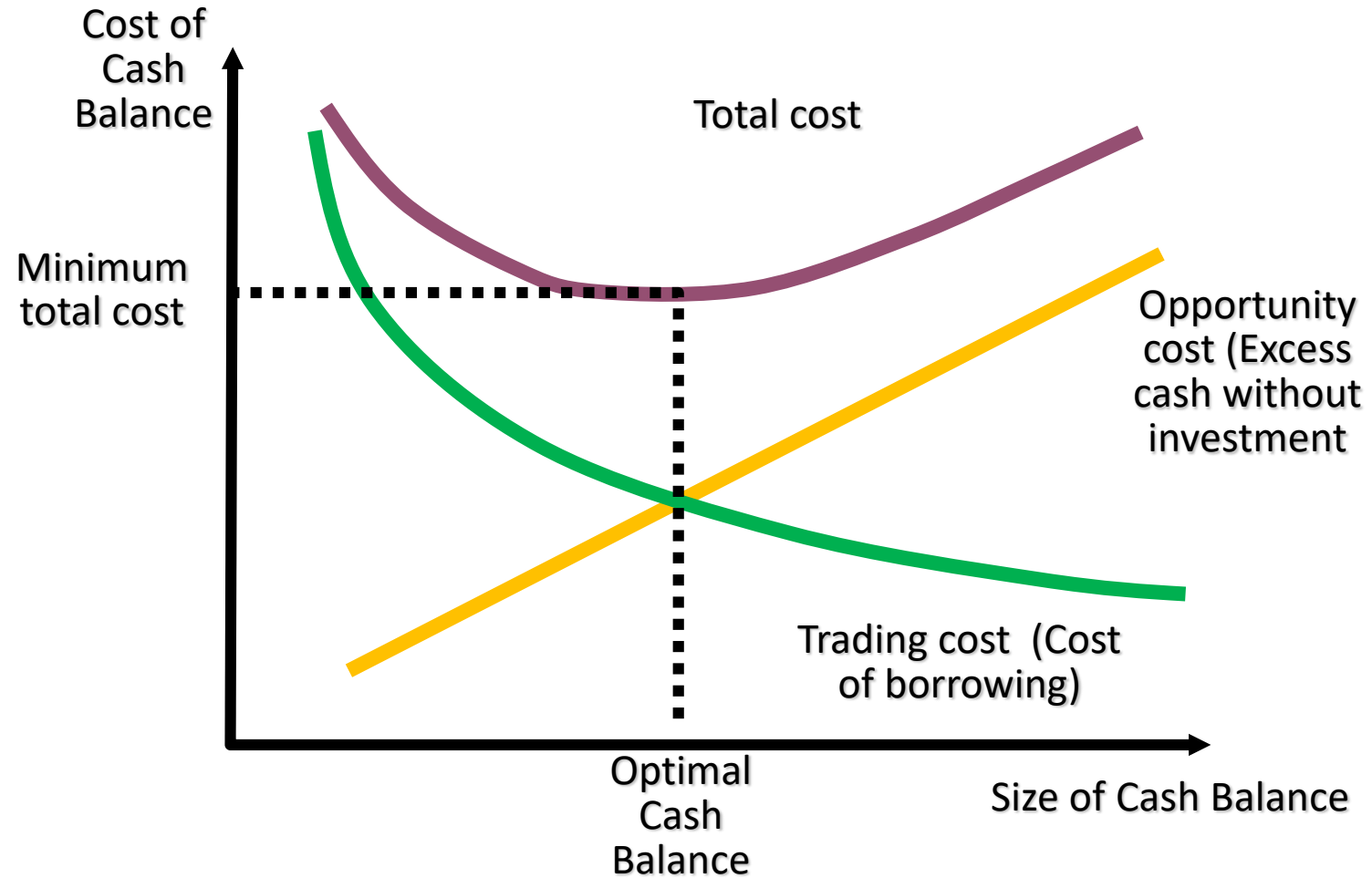


Cash Management Model

- Target cash balance aims at minimizing the opportunity cost of holding additional cash and the cost of holding too little cash
- Flexible working capital policy will invest additional working capital cash in marketable securities. These funds tied up in investments will increase the cost of unforeseen shortage for cash payments. The shortage costs will be costs associated with buying and selling securities.

Optimal Cash Balance

- Trade-off Costs for Determining Optimal Cash Balance



Baumol-Allais-Tobin Model (BAT)

- Transactions costs (brokerage costs of selling marketable securities) occur when cash is replenished, implying large initial balances will lower the trading costs connected with cash management. However, the larger the average cash balance, the greater is the opportunity cost (return that could have been earned on marketable securities)
- F = Fixed cost of trading securities to replenish cash.
- T = Total amount of new cash needed for transactions purposes over the relevant planning period (one year).
- R = Opportunity cost of holding cash (interest rate on marketable securities).
- $T = \text{Opportunity cost} + \text{Trading Cost} = [(C/2) \times R] + [(T/C) \times F]$

Where: Opportunity cost = Average cash x opportunity cost = $(C/2) \times R$

Trading cost = Securities to be sold x fixed cost of trading = (Total cash need / Initial cash balance) x Fixed trading cost = $(T/C) \times F$

- Optimal cash balance: Opportunity cost = Trading cost
- $C^2 = (2T \times F) / R$: $C = \sqrt{(2T \times F) / R}$

BAT Model: Example

- Opportunity Cost: $(C/2) \times R$

Month	Initial Cash Balance	Average Cash Balance	Opportunity Cost (R =10%)
1	4,800,000	2,400,000	240,000
2	2,400,000	1,200,000	120,000
3	1,200,000	600,000	60,000
4	600,000	300,000	30,000
5	300,000	150,000	15,000
6.....			

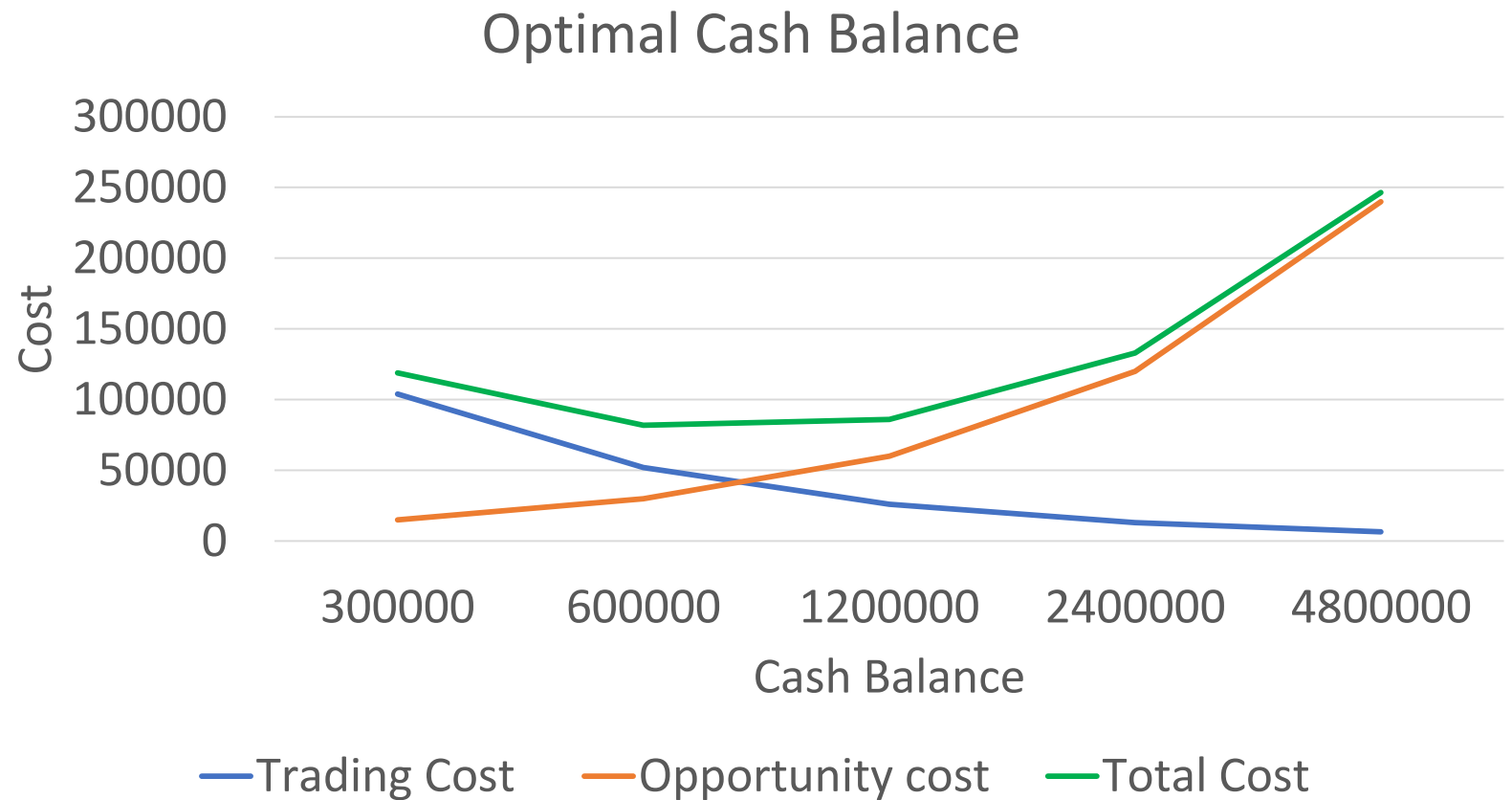
BAT Model: Example

- Trading Costs: $(T/C) \times F$

Month	Total Disbursements	Initial Cash Balance	Trading Cost (F=1,000) $(T/C) \times F$
1	31,200,000	4,800,000	6,500
2	31,200,000	2,400,000	13,000
3	31,200,000	1,200,000	26,000
4	31,200,000	600,000	52,000
5	31,200,000	300,000	1,04,000
6.....			

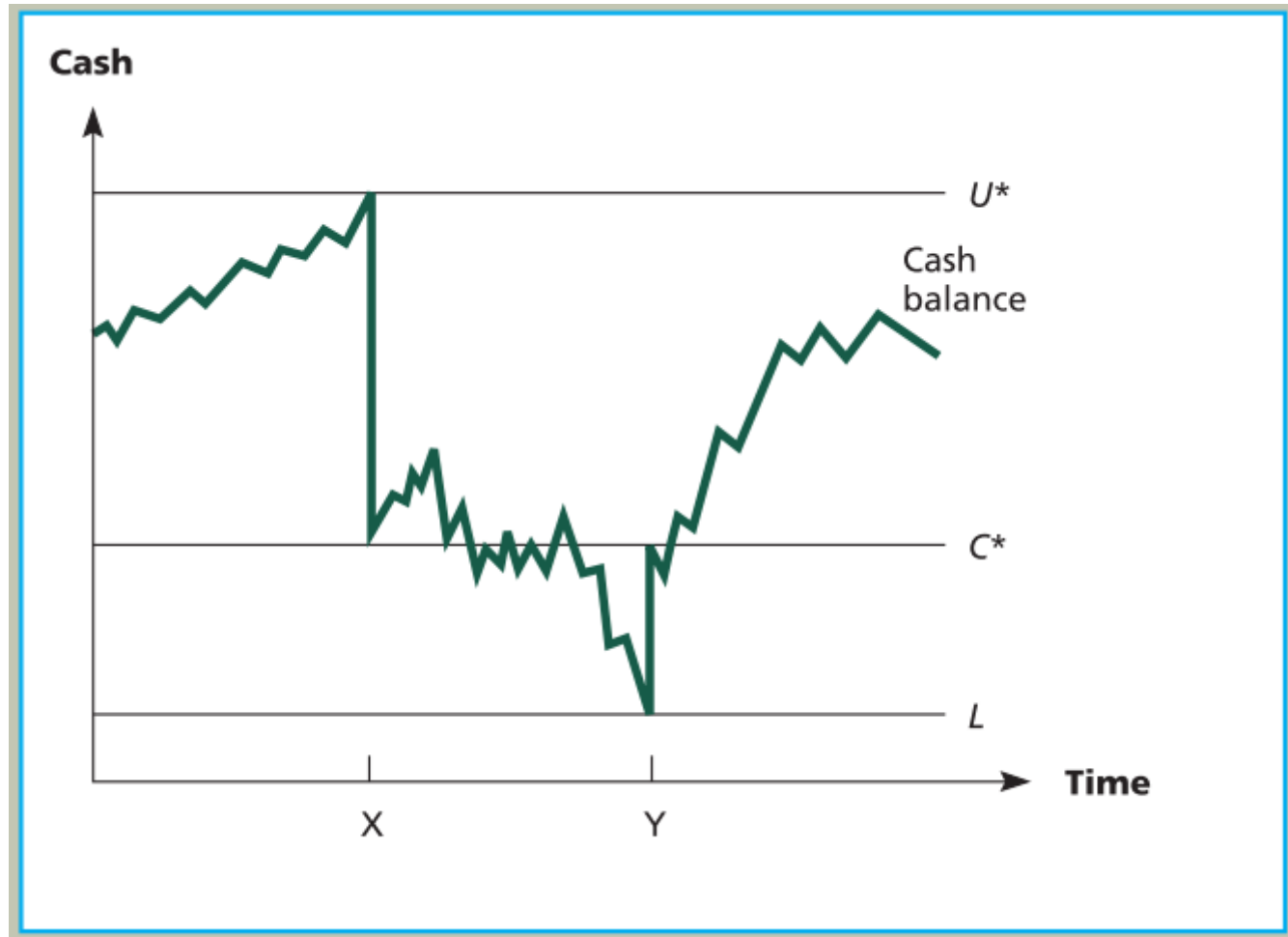
BAT Model: Example

- Optimal Cash Balance
- $C = \sqrt{(2T \times F)/R}$
- $C = 790,000$



Miller-Orr Model

- U^* = Upper Control Limit
- L = Lower Control Limit
- C^* = Target Cash Balance
- No Transaction: Cash Balance Between C^* and U^*
- Sell Securities: Cash Balance below C^* ($C^* - L$)
- Buy Securities: Cash Balance Above U^* ($U^* - C^*$)



Miller-Orr Model

- Set the lower control limit (L)
- Estimate the standard deviation of daily cash flows σ
- Determine the effective interest rate
- Estimate trading costs of buying and selling securities
- Best return point (C^*) is increasing in trading costs (F) and decreasing in opportunity cost (R)
- C^* and C (average cash balance) are positively related to the variability of cash flows

Average Cash Flows

- When cash flows have the model estimate, average cash flows are:

$$\frac{4C - L}{3}$$

Miller-Orr Model

- Expected balances replace the average balance
- Firm sets the lower limit (L), then the target cash level and the upper boundary

$$C^* = L + \sqrt{\frac{3F\sigma^2}{4R}}$$

$$U^* = 3C^* - 2L$$

Miller-Orr Model: Example

- Assume annual opportunity cost $R=10\%$, and Fixed cost $F=1,000$.
- Assume σ , the standard deviation of daily cash flows: 2,000
The variance (σ^2) is then 4,000,000
- Effective Annual Rate for 10% compounded daily, gives a daily opportunity cost.

$$\sqrt[365]{1.10} - 1 = 0.000261 = R$$

- Assume $L = 0$

$$C^* = 0 + \sqrt[3]{\frac{3 \times 1,000 \times 4,000,000}{4 \times 0.000261}} = 22,568$$

$$U^* = 3 \times 22,568 = 67,704$$